La Scuola d'Italia / Dr. Huson / IB Math: Review
18 March 2026

11.8 Homework: Mixed Review

— Section A: No Calculator —

1. [Maximum mark: 5]

   An arithmetic sequence has $u_5 = 17$ and $u_{12} = 38$.

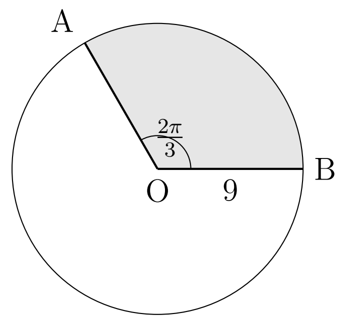
   (a) Find the common difference d. [2]

   (b) Find the first term u_1. [1]

   (c) Find the sum of the first 20 terms, S_{20}. [2]

2. [Maximum mark: 5]

The following diagram shows a circle with centre O and radius 9 cm. Points A and B lie on the circumference. The angle $\hat{AOB} = \dfrac{2\pi}{3}$ radians.



(a) Find the exact length of arc AB. [2]

(b) Find the exact area of the shaded sector. [3]

3. [Maximum mark: 5]

Consider the function $f(x) = 2x^3 - x^2 + 3$.

(a) Find $f'(x)$. [2]

(b) Find the equation of the tangent to the graph of f at $x = 1$. [3]

— Section B: Calculator Allowed —

4. [Maximum mark: 5]

The following table shows the number of hours of study (x) and the test score (y) for seven students.

Hours (x)	2	3	4	5	6	7	8
Score (y)	48	55	58	70	74	82	85

The regression line of y on x can be written as $y = ax + b$.

(a) Find the value of a and the value of b. [2]

(b) Write down the value of Pearson's product-moment correlation coefficient, r. [1]

(c) Use your regression line to estimate the test score for a student who studied for 5.5 hours. [2]

5. [Maximum mark: 6]

Triangle ABC has $b = 7\,\text{cm}$, $c = 10\,\text{cm}$, and $\text{B}\hat{\text{A}}\text{C} = 52°$.

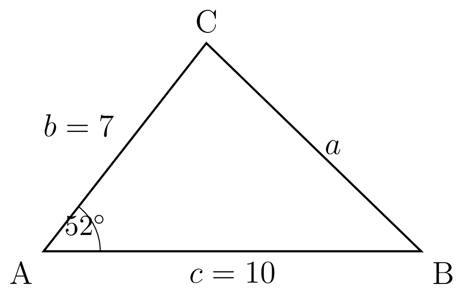


diagram not to scale

(a) Find the length of side a. [3]

(b) Find the area of triangle ABC. [3]

6. [Maximum mark: 5]

In triangle PQR, $Q\hat{P}R = 40°$, $P\hat{R}Q = 75°$, and $p = 8\,\text{cm}$.

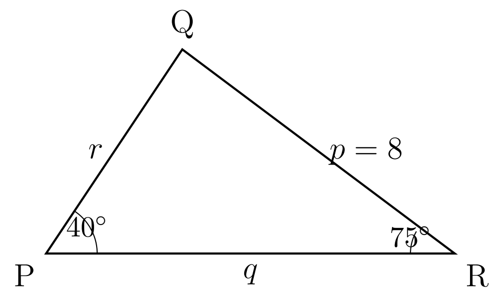


diagram not to scale

(a) Find the measure of angle $P\hat{Q}R$. [1]

(b) Find the length of side q. [2]

(c) Find the area of triangle PQR. [2]

7. [Maximum mark: 6]

The weights of apples from an orchard are normally distributed with mean 150 g and standard deviation 12 g.

(a) Find the probability that a randomly chosen apple weighs less than 165 g. [2]

(b) Find the probability that a randomly chosen apple weighs between 140 g and 170 g. [2]

(c) The heaviest 5% of apples are classified as "jumbo". Find the minimum weight for a jumbo apple. [2]

8. [Maximum mark: 5]

Olivia invests \$5000 in a savings account that pays an annual interest rate of 3.2%, compounded quarterly.

(a) Find the value of the investment after 6 years. [3]

(b) Determine how many full years it takes for the investment to double in value. [2]

9. [Maximum mark: 5]

   Events A and B are such that $\mathrm{P}(A) = 0.6$, $\mathrm{P}(B) = 0.5$, and $\mathrm{P}(A \cup B) = 0.85$.

   (a) Find $\mathrm{P}(A \cap B)$. [2]

   (b) Find $\mathrm{P}(A \mid B)$. [1]

   (c) Determine whether events A and B are independent. Justify your answer. [2]

10. [Maximum mark: 5]

A factory produces light bulbs, and 8% of them are defective. A quality inspector selects 20 bulbs at random. Let X be the number of defective bulbs in the sample.

(a) State the distribution of X. [1]

(b) Find $\mathrm{P}(X = 2)$. [1]

(c) Find $\mathrm{P}(X \geq 3)$. [2]

(d) Find the expected number of defective bulbs. [1]